Educating for the AI Revolution:
Fostering a Holistic Human-Centered Approach Based on Wisdom and Multiplexity

JUNAID QADIR
Professor of Computer Engineering,
CSE Department, CENG,
Qatar University



HBKU CIS Bridging Frontiers (21 Feb 2024)

Outline

1. The Generative AI Disruption
2. Role of Wisdom and Multiplexity
3. Education for the Exponential AI Era

1

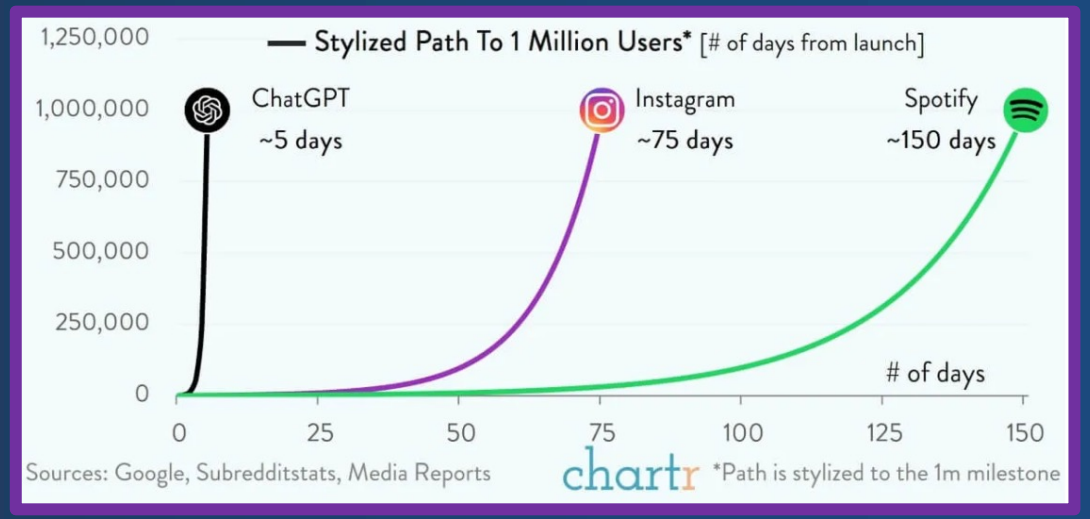
The Generative AI Disruption

Image Generator

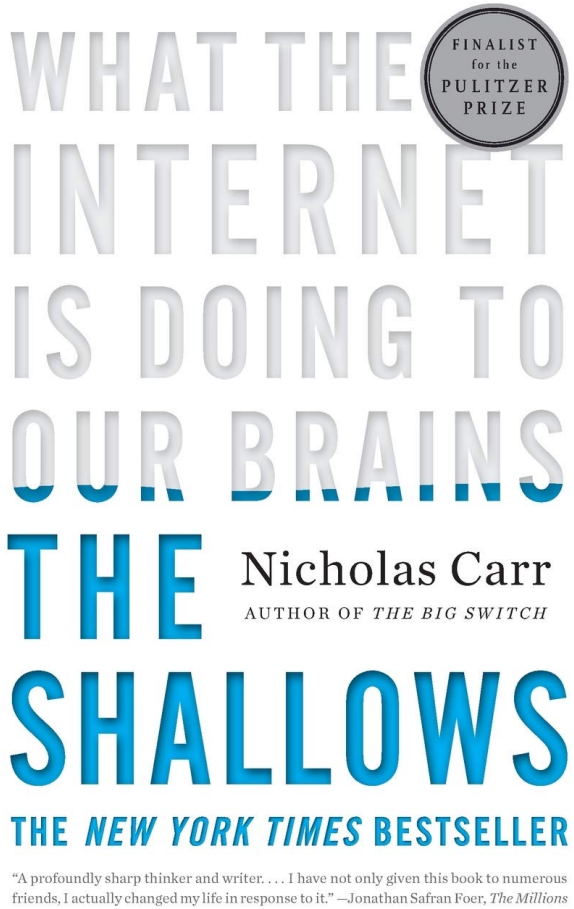
Speech Synthesizer

Text Generator

Conversational AI (ChatBot)

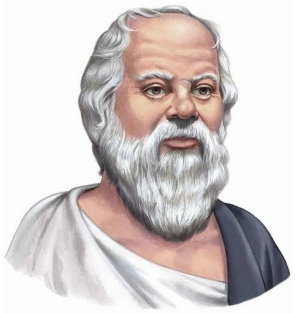


The risk of outsourcing thinking to technology



Plato's Argument Against Writing

"If men learn this, it will implant forgetfulness in their souls. ***They will cease to exercise memory because they rely on that which is written, calling things to remembrance no longer from within themselves, but by means of external marks.***"

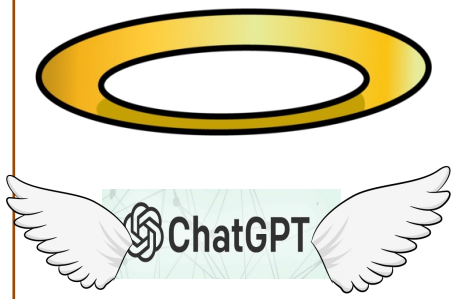


Such tools impart "**not truth but only the semblance of truth" and that those who adopt it will appear to omniscient and will generally know nothing**."

How we may go wrong with GenAI for Education

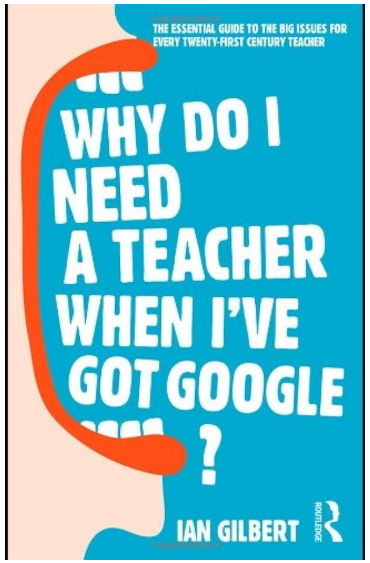
HALO EFFECT

The Gullibility to Believe AI is infallible



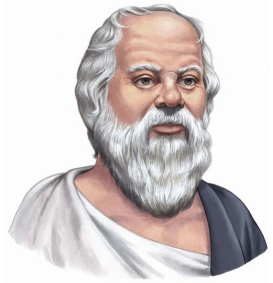
DISINCENTIVES

Why spend time and effort learning when you can just Google/ or use ChatGPT



SOPHISTRY & PRETENTION

Appearances become more important than truth. In Education, the law of harvest applies.





TECHNOLOGICAL OUTSOURCING



AL-GHAZALI AND THE ROBBERS

DEHUMANIZATION

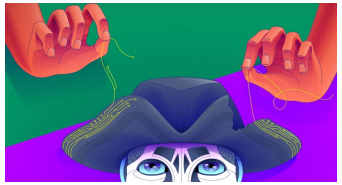
Making the thoughts/ outputs of every student formulaic and cliched



How we may go wrong with GenAI for Education

AI COLONIALISM

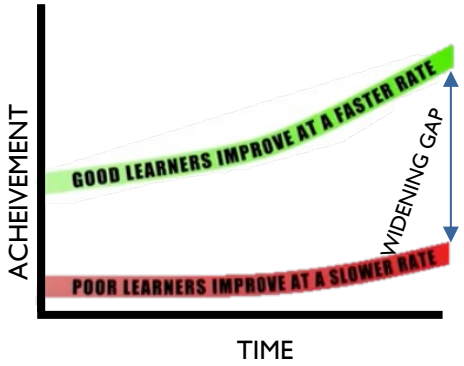
The "Global North"—the rich dominating countries stand to dominate and monopolize the market



MARGINALIZING THE MARGINALIZED
Data-poor regions, not having indigenous LLMs, are put at a long-term risk of being colonized by standards embedded in LLMs

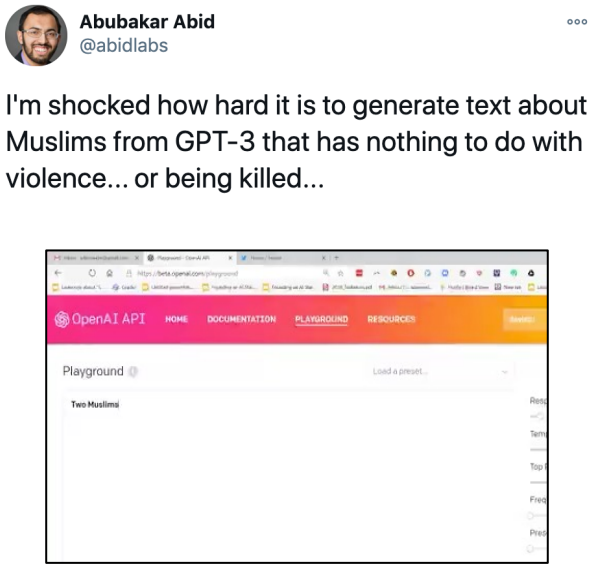
THE GENAI EDTECH MATTHEW EFFECT

Those who already know, stand to benefit most from GenAI, as they can prompt, critique and judge the output better



REINFORCING BIAS

Those who already know, stand to benefit most from GenAI, as they can prompt, critique and judge the output better



2

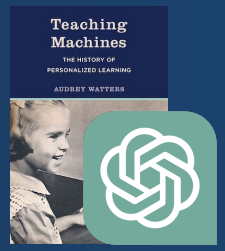
Role of Wisdom and Multiplexity

MACHINE-CENTRIC

THE TURING TRAP

CAN MACHINES THINK?





AUTOMATION/ REPLACEMENT FOCUS

THE TEACHING MACHINE TRAP

CAN MACHINES TEACH?

Note the exclusive focus on machine intelligence
(without considering effects on humans)

Social systems as complex adaptive systems



TECHNO-SOLUTIONISM:

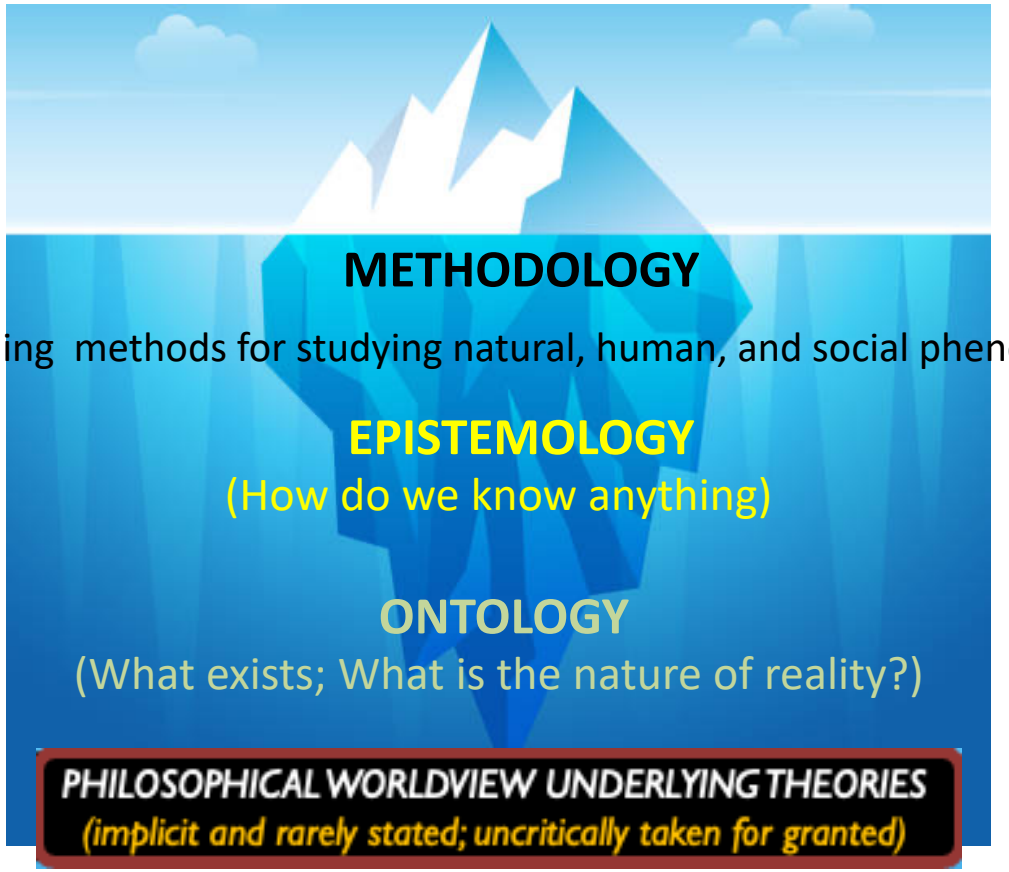
"For every complex problem there is an answer that is clear, simple, and wrong."—H. L. Mencken

"The chief cause of problems is solutions."—Sevareid

"No problem stays solved in a dynamic environment."—*Russell Ackoff*

Resisting reductionism and uniplexity

Uniplex Approaches
Uses a reductionist impoverished worldview



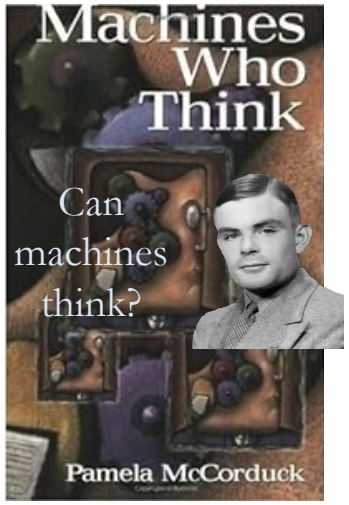
Multiplex Approaches
Uses a multi-layered comprehensive approach for explaining reality

Şentürk, Recep et al. *Comparative theories and methods between uniplexity and multiplexity*, 2020.

Technology emerges from a worldview

OUR NOTIONS OF EDUCATION IS INFLUENCED BY HOW WE UNDERSTAND HUMANS AND PROGRESS

Flip sides of the same (machine-centric) worldview:
(1) thinking machines and *(2) robotic humans*



I, RODNEY BROOKS, AM A ROBOT



I am a machine. So are you.

Of all the hypotheses I've held during my 30-year career, this one in particular has been central to my research in robotics and artificial intelligence. I, you, our family, friends, and dogs—we all are machines. We are really sophisticated machines made up of billions and billions of biomolecules



Homo economicus

The term ***homo economicus***, or **economic man**, is the portrayal of humans as agents who are consistently rational, narrowly self-interested, and who pursue their subjectively-defined ends optimally.

Consider economics (an example social science) in which human beings are considered selfish greedy "rational" mechanical utility maximizers.

METAPHORS SHAPE THE WAY YOU THINK

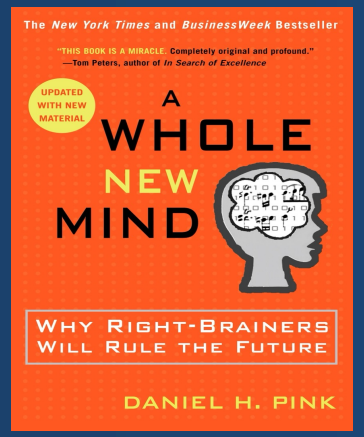
"Metaphorical thinking half discovers and half invents the likenesses it describes."—James Geary

3

Education for the Exponential AI Era

Developing AI-Proof Learners





Holistic Human-Centered View of AI for Education

(focused on what the AI can support humans in)

THE CHARACTER IMPERATIVE



THE ECONOMIC IMPERATIVE

THE UNIQUE HUMAN POTENTIAL REALIZATION IMPERATIVE

INFORMED JUDGEMENT AND CITIZENSHIP

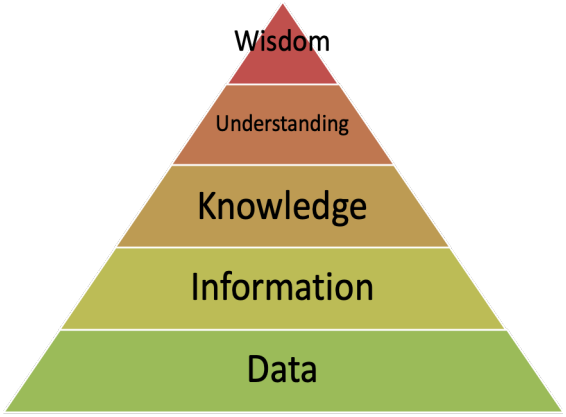
ENGENDERING SHARED PROSPERITY

First Law
Your work should be no worse than that produced by the AI alone.
(Embrace change and leverage AI)

Second Law
(Corollary of Law 1)
If you don't want to be made irrelevant, maintain your competitive advantage
(Become **AI Proof** and maintain some competitive advantage)

Third Law
After becoming users of AI, don't let your inherent unaided capabilities drop
(Don't outsource your thinking muscle allowing it to atrophy]

What would educating for wisdom look like?



Wisdom emerges from an understanding of the historical, social, and cultural context. It cannot be completely programmed into logic, mathematics, models, or code

CRITICAL THINKING

—Knowing the context

—Knowing the pitfalls of rules/model/methods

—Ability to engage in informed judgement

—Knowing the limits of your knowledge

الجهل البسيط

الجهل المركّب

DUNNING-KRUGER

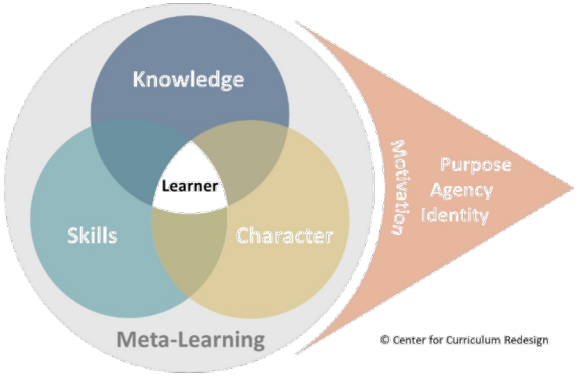
HOLISTIC THINKING

—Multidimensional/ Multidisciplinary Thinking

—Apart from Cognitive, Focus on Character, Skills, & Metacognition

—Intelligence requires both reckoning & judgement

—Dialogic Thinking

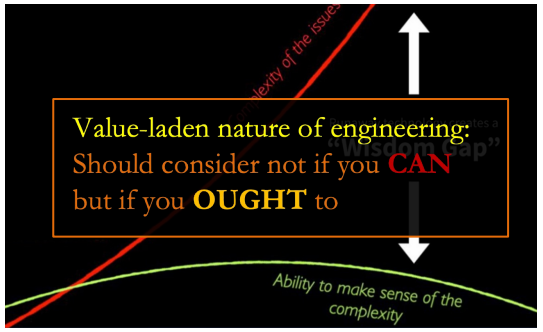


PURPOSEFUL USE OF AI

— Knowing which ends/ objectives/ purpose to pursue and prioritize

— Importance of Human Skills and Character

—Pro-Human & Pro-Social



Conclusion:

Our aim in education should focus on creating the following awareness in our students:

1. *how they can use AI as a tool that contributes to human flourishing & shared prosperity;*

2. *how they can take actions as individuals and society to avoid its potential harms (such as AI colonialism; dehumanization)*

The focus should be on what the humans can do with and without the technology and not on what the algorithms can do on their own.

THANK YOU!

Your feedback is most welcome at

jqadir@qu.edu.qa